

Enterprise AI Knowledge OS: System Architecture

ARCHITECTURE BLUEPRINT & HIGH-LEVEL SYSTEM DESIGN

Executive Summary

The Enterprise AI Knowledge OS unifies unstructured organizational documentation, structured relational data, and real-time semantic search into an enterprise-grade retrieval pipeline.

Core Architectural Layers

1. Ingestion Layer (PDF/JSON/CSV parsers)
2. Embedding & Vector Indexing Engine (FAISS + Dense Models)
3. Orchestration & Retrieval Engine (RAG + Context Fusion)
4. Security & Access Control (RBAC Filter Layer)

Scalability & Resilience

Horizontal scaling of vector shards, fallback caching layer for frequently asked semantic queries, and strict end-to-end trace logging.